Unit 2 – Selection and Iteration Name: _____
Circuit Activity – Tracing if Statements

Beginning in cell #1, trace through the code to determine what is printed. Once you see what is printed, hunt for your answer and make that problem #2. Continue in this manner until you complete the circuit.

Answer: 5, –9

#1

```
int x = 10;
int y = 5;
if(x / y == 2)
{
    y += 3;
    if(y % 2 == 0)
    {
        x -= 4;
    }
}
System.out.println(x + ", " + y);
```

Answer: 7, 4

_____

```
double a = 6.4;
double b = -3.2;
if(Math.abs(b) > a)
{
    a = Math.pow(a, 2);
}
else if(Math.sqrt(a) == 2.5)
{
    b *= -1;
}
System.out.println((int) a + ", " +
(int) b);
```

Answer: 9, 8

_____

```
boolean flag = true;
int num = 15;
int val = 3;
if(flag && num % 3 == 0) {
    num /= 5;
    val += 3;
}
else if(!flag || num > 10) {
    num += 2;
    val += 2;
}
System.out.println(num + ", " + val);
```

Answer: 3, 7

_____

```
int x = 12;
int y = 4;
if(x > y && y % 2 == 1 || x > 10)
{
    x /= y;
}
if(x * y < 10)
{
    y += 6;
}
else
{
    x -= 2;
}
System.out.println(x + ", " + y);
```

Answer: 8, 6

#_____

```
int a = 5, b = 3, c = 7;
if(a + b > c && b * c < 25) {
    b -= a;
}
else {
    a += b;
}
if(a + b + c > 10){
    a += c;
}
else {
    b -= c;
}
System.out.println(a + ", " + b);
```

Answer: 6, –3

#_____

```
int num = 9;
int val = 2;
if(num % val == 0) {
    num = num / val;
    val = 3;
} else if(num % val == 1) {
    num = num * 3;
    val = 4;
} else {
    num += 10;
    val = 5;
}
System.out.println(num + ", " + val);
```

Answer: 6, 8

#_____

```
int a = 8, b = 3;
if(a % b == 2) {
    a = (int) Math.pow(b, 2);
    if(a % b == 0)
        b = (int) Math.pow(2, b);
}
else if (b + a > 10) {
    b *= 2;
    if(b == 6)
        a = (int) Math.pow(a, 2);
}
else {
    a -= b;
}
System.out.print(a + ", " + b);
```

Answer: 27, 4

#_____

```
int s = 12, t = 5;
if(s / t == 2) {
    t += s % t;
    if(t > 6) {
        s -= 3;
    }
}
else {
    s += 2;
}
System.out.print(s + ", " + t);
```

Answer: 9, 14

#_____

```
int a = -4, b = 7;
if(Math.abs(a - b) < 5) {
    a *= 2;
    if (a + b > 15)
        b -= 3;
}
else {
    a += b;
}
System.out.print(a + ", " + b);
```

Answer: 10, 4

#_____

```
int c = 9, d = 4;
if(c != 0 && c % d == 1) {
    c += 2;
    if(c / d > 2) {
        d -= 1;
    }
}
if(c % d == 1 || c != 0){
    c -= d;
}
System.out.print(c + ", " + d);
```

Answer: 9, 7

_____

```
int a = 10;
int b = 2;
if(a % b == 0 && a / b > 4) {
    a -= 6;
    if (b * 3 < a) {
        b += 4;
    }
}
if(a / b >= 2 || a < b) {
    a += b;
    if(a == b) {
        b += 1;
    }
}
System.out.print(a + ", " + b);
```

Answer: 1, 4

_____

```
int w = 12, x = 5;
if(w % 3 == 0) {
    if(x > 4) {
        w -= 6;
        if(w % 2 == 0) {
            x += 3;
        }
        else {
            x -= 1;
        }
    }
    else {
        x += 2;
    }
}
else {
    w += 4;
    if(x < 10) {
        x *= 2;
    }
}
System.out.print(x + ", " + w);
```

Answer: 6, 2

_____

```
int ans = 7, count = 14;
if(count % ans == 0) {
    ans += 2;
    if(count % 4 == 0) {
        count /= 2;
        if(ans * count > 20) {
            ans -= 5;
        }
        else {
            count += 3;
        }
    }
}
else {
    count -= 7;
    if(count > 0)
        ans *= 2;
}
System.out.print(ans + ", " + count);
```

Answer: 3, 6

_____

```
int x = 20, z = 6;
if(Math.sqrt(x) > 4) {
    if(z % 3 == 0) {
        x /= 2;
        if(x % 5 == 0) {
            z -= 2;
        }
    }
    else {
        x += 3;
    }
}
else {
    z += 5;
    if(x - z < 10) {
        x *= 2;
    }
}
System.out.print(x + ", " + z);
```